TRUECHAIN – AUTHENTIC PRODUCT TRACKING USING BLOCKCHAIN ACROSS SUPPLY CHAIN

Submitted in partial fulfilment of the requirements of the degree of

BACHELOR OF COMPUTER ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the project entitled "**TrueChain – Authentic Product Tracking using Blockchain across Supply Chain**" is a bonafide work of **Aman Singh (22102076), Nishant Singh (22102031), Gaurav J. Shah (21102121), and Rohan Rathod (21102122)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Engineering.**

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Project Report Approval for B.E.

This project report entitled **"TrueChain – Authentic Product Tracking using Blockchain across Supply Chain"** by **Aman Singh, Nishant Singh, Gaurav J. Shah, and Rohan Rathod** is approved for the degree of **Bachelor of Engineering** in **Computer Engineering**, **2025-2026**.

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2. _____	_____

Date:

Place:

Declaration

We declare that this written submission represents our ideas in our own words andwhere others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that We have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly citedor from whom proper permission has not been taken when needed.

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Abstract

The rise of counterfeit products poses a serious challenge to luxury brands, consumers, and global markets. From watches and handbags to electronics, the increasing sophistication of counterfeiters makes traditional verification methods like certificates or holograms unreliable. The True Chain project addresses this issue by developing a blockchain-based product authentication system that enables secure, transparent, and real-time tracking of goods across the entire supply chain. Each product is assigned a unique QR code that connects to a blockchain ledger, recording every stage from manufacturing to final sale. The system ensures immutability, meaning once data is entered, it cannot be altered or deleted. Smart contracts automate verification and ownership transfers, while blockchain's decentralized structure eliminates the need for a central authority. Built using React.js, Node.js, Solidity, Hardhat, and PostgreSQL, True Chain provides a scalable infrastructure that improves supply chain transparency, consumer trust, and brand protection. By merging blockchain and QR technology, True Chain establishes a tamper-proof digital ecosystem that not only prevents counterfeit goods but also promotes responsible manufacturing and sustainable business growth. The project aligns with SDG 8 – Decent Work and Economic Growth, SDG 9 – Industry, Innovation, and Infrastructure, SDG 12 – Responsible Consumption and Production promoting secure trade practices, product authenticity, and sustainable business growth through digital innovation.

Keywords: - *Blockchain, Luxury Products, Counterfeit Detection, QR Code, Smart Contracts, Authentication, Transparency, Brand Protection*.

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ABBREVIATION

DApp	Decentralized Application
QR	Quick Response
IPFS	Inter Planetary File System
SDG	Sustainable Development Goals
UI	User Interface
API	Application Programming Interface
DB	Database
UML	Unified Modeling Language
DFD	Data Flow Diagram
DAO	Decentralized Autonomous Organization
AI	Artificial Intelligence
CRUD	Create, Read, Update, Delete
SDK	Software Development Kit
NFT	Non-Fungible Token
JSON	JavaScript Object Notation
ERC	Ethereum Request for Comment
ETH	Ethereum (Cryptocurrency)
DBMS	Database Management System
SDLC	Software Development Life Cycle
JWT	JSON Web Token
CI/CD	Continuous Integration / Continuous Deployment

CHAPTER 1

Introduction

In today's competitive global economy, luxury brands are facing an alarming threat from counterfeit products. The problem not only leads to financial losses but also damages brand image, reduces consumer trust, and undermines legitimate market operations. Traditional verification techniques such as certificates, holographic tags, or serial numbers are no longer sufficient. These physical identifiers can be easily copied or tampered with by counterfeiters, allowing fake products to circulate in the market without detection. True Chain provides a modern, technology-driven solution to this problem by combining Blockchain and QR Code technology to create a transparent, tamper-proof, and decentralized product tracking system.

Blockchain technology stores all product data in an immutable ledger that cannot be altered. Each product, when manufactured, is registered on the blockchain network and assigned a unique digital identity represented by a QR code. As the product moves through the supply chain — manufacturer → distributor → retailer → customer — each transaction or ownership change is recorded on the blockchain, ensuring end-to-end traceability.

Consumers can scan the QR code at any time using a smartphone to verify product authenticity and view its complete journey. This creates an ecosystem of trust, transparency, and accountability, reducing counterfeit circulation and protecting both businesses and consumers.

Key features of the project:

- Each luxury product is registered on the blockchain with a unique, tamper-proof digital identity linked to a QR code.
- Smart contracts automate verification and ownership transfer, ensuring accuracy and transparency.
- QR code scanning lets consumers instantly check product authenticity and trace its full journey.
- The system enhances brand protection, consumer trust, and prevents counterfeit circulation.
- Supports ethical sourcing and sustainability by verifying material origins through blockchain records.

The TrueChain system **combines** blockchain for security**,** QR codes for transparency**,** and smart contracts for automation, ensuring luxury product data remains secure, traceable, and tamper-proof. This decentralized approach reduces counterfeit risks, improves supply chain efficiency, and builds trust among brands and consumers. Blockchain provides an immutable record of each product's journey from manufacturing to sale, while smart contracts automate verification and ownership updates.

Aligned with the **UN Sustainable Development Goals**, True Chain supports SDG 8 – Decent Work and Economic Growth**,** SDG 9 – Industry, Innovation, and Infrastructure, and SDG 12 – Responsible Consumption and Production**,** promoting ethical trade, transparency, and sustainable growth in the luxury industry.

CHAPTER 2

Literature Survey

Recent studies highlight the growing use of blockchain and QR code technology in combating counterfeit products and improving supply chain transparency.

1. Anjum & Dutta (2022) proposed an Ethereum-based decentralized application (DApp) that uses QR codes for real-time product verification. Their system showed that blockchain can effectively manage ownership tracking and confirm authenticity.
2. Indiana Publications (2021) demonstrated how integrating blockchain with IoT can strengthen counterfeit detection by using connected devices to improve tracking precision.
3. Kshetri (2018) discussed blockchain's tamper-proof nature, emphasizing how immutable ledgers prevent data manipulation and enhance product traceability.
4. IEEE (2021) presented a blockchain-driven anti-counterfeit model using dynamic QR codes that update automatically to reflect real-time product status, reducing the risk of duplication. Springer (2024) implemented a blockchain-based product authentication system linked with Ethereum smart contracts, allowing instant verification of provenance and transaction history by all stakeholders.
5. Finally, OA Journal (2025) explored practical use cases combining blockchain and QR code scanning across industries, proving their effectiveness in detecting and preventing counterfeit goods.

Sr. No.	Paper Name	Publisher, Year	Strengths	Drawback
[1]	Identifying Counterfeit Products using Blockchain Technology in Supply Chain System – N. Anjum & P. Dutta	arXiv, 2022	Ethereum-based DApp uses QR codes for real-time ownership verification and authentication.	Limited scalability and high transaction costs due to Ethereum gas fees
[2]	Detecting Counterfeit Products in Supply Chains via Blockchain–IoT Integration	Indiana Publications, 2021	Combines IoT and blockchain for real-time tracking, improving traceability accuracy.	IoT device dependency increases cost and integration complexity
[3]	Blockchain Solution for Counterfeit Detection	IEEE, 2021	Uses blockchain for quality assurance and fraud prevention in supply chains.	Focused on simulation, lacks real-world implementation and testing.
[4]	Blockchain-Driven Anti-Counterfeit Mechanism with Dynamic QR Codes	International Journal for Multidisciplinary Research, 2024	Introduces dynamic QR codes that update in real time to prevent cloning.	Requires continuous internet access and complex QR code synchronization.

[5]	Product Authentication and Traceability Using Blockchain	Springer, 2024	Links blockchain with Ethereum smart contracts for instant verification and provenance tracking.	High computational and storage cost limits scalability for large datasets.
[6]	Blockchain: A Secure and Transparent Solution to Detect Counterfeit Products	Springer Professional, 2021	Ensures certificate validation and provenance tracking through blockchain and QR codes.	Limited user interface and no integration with consumer-facing applications.
[7]	Implementation of Leveraging QR Codes and Blockchain	OA Journal of Innovative Science & Engineering, 2025	Real-world use case combining blockchain and QR scanning for product authentication.	Depends heavily on network reliability for live verification.
[8]	Decentralizing Supply Chain Anti-Counterfeiting Systems Using Blockchain Technology	arXiv, 2021	Uses NFC and blockchain for tamper-proof authentication and traceability.	NFC integration increases production cost for small-scale industries.
[9]	Chain Thread: Blockchain-Powered Product Verification and Authenticity Tracking	IJRASET, 2024	Consumer-friendly DApp with QR-based product validation and smart contract automation.	Limited security evaluation and lacks performance testing at scale.
[10]	Identifying Fake Products through a Barcode-Based Blockchain System	ResearchGate / IEEE-style, 2024	Uses blockchain ledger with barcode or QR scan for provenance and authenticity.	Barcode systems less secure than dynamic QR or NFC solutions.

[11]	Blockchain for Provenance Tracking in Luxury Goods	MDPI, 2023	Focuses on luxury supply chains, ensuring transparency and anti-counterfeit measures.	Still in early prototype stage; lacks integration with retail systems.
[12]	Smart Contract-Based Traceability for Supply Chains	Elsevier, 2022	Introduces automated verification using Solidity smart contracts to enhance transparency.	Requires technical knowledge for maintenance and contract deployment
[13]	Anti-Counterfeit Framework Using Ethereum Blockchain and IPFS	IEEE Access, 2022	Integrates blockchain and IPFS for secure and decentralized product data storage.	Storage and retrieval times are slow for large multimedia data.
[14]	Blockchain-Powered Authenticity Verification in E-Commerce	Springer, 2023	Improves online retail transparency by validating seller and product authenticity.	Security issues persist in linking off-chain product data with blockchain.
[15]	Blockchain and QR Technology for Supply Chain Security	ACM Digital Library, 2024	Highlights real-time verification, traceability, and transparency using hybrid blockchain.	Limited consensus algorithm efficiency for large-scale enterprise deployment.

Table 2.1: Literature Survey Table

CHAPTER 3

Limitation of Existing System

Traditional product authentication and supply chain systems, which rely on paper-based certificates, holograms, barcodes, or centralized databases, face multiple challenges compared to a blockchain-based system like **TrueChain**.

Key limitations of existing systems:

- **Lack of Transparency and Traceability:** Conventional systems do not provide end-to-end visibility of a product's journey. Once a product leaves the manufacturer, tracking its movement through distributors and retailers becomes difficult, allowing counterfeit goods to enter the supply chain undetected.

- **Data Tampering and Forgery:** Centralized databases and physical identifiers such as barcodes or holograms can be easily duplicated, altered, or forged. This compromises the authenticity of product information and enables counterfeiters to manipulate records.

- **Absence of Real-Time Verification:** Existing methods often require manual verification through certificates or labels, which can be lost, outdated, or faked. There is no instant way for consumers to verify product authenticity directly at the point of purchase.

- **Limited Consumer Awareness and Trust:** Consumers have minimal access to reliable authenticity data. Without transparent verification mechanisms, they must rely on brand reputation or seller assurances, which are often misleading in counterfeit-prone markets.

- **Dependency on Centralized Authorities:** Most systems depend on a central body or intermediary for product verification, creating single points of failure and making the process vulnerable to corruption, data breaches, or insider manipulation.

- **No Immutable Record Keeping:** Traditional systems do not maintain tamper-proof logs of product transactions, leading to disputes, lack of accountability, and difficulty in tracing counterfeit origins.

In summary, current product verification systems lack transparency, real-time verification, and security. These limitations underline the need for an advanced, decentralized, and immutable approach like **TrueChain**, which combines blockchain and QR technology to ensure trust, traceability, and authenticity throughout the supply chain.

CHAPTER 4

Problem Statement, Objectives and Scope

4.1 Problem Statement

The global market for luxury goods, electronics, and fashion faces a continuous battle against counterfeit products. Despite advancements in packaging and tracking technologies, counterfeiters exploit loopholes in centralized systems, producing fake products that appear genuine. The absence of a decentralized, tamper-proof system for product verification allows fake goods to infiltrate the supply chain undetected. Traditional methods like barcodes and holograms can be forged or reused. This creates a pressing need for an innovative, transparent, and secure solution that ensures authenticity and traceability of products from manufacturing to the end consumer.

4.2 Objectives

- **Integrate Blockchain for Authentication:** Implement a blockchain-based framework that securely records all product data and transactions in an immutable ledger.
- **Enable Instant Verification:** Provide users with a quick and easy way to verify product authenticity using QR code scanning.
- **Build Consumer Trust:** Offer complete product transparency to boost customer confidence and brand loyalty.
- **Streamline Supply Chain Processes:** Improve efficiency and coordination among manufacturers, suppliers, retailers, and consumers.
- **Promote Sustainable and Ethical Sourcing:** Allow brands to trace the origin of raw materials and ensure responsible production practices.
- **Enhance Data Integrity and Security:** Use blockchain cryptography to protect data from unauthorized modification or loss.

4.3 Scope

The scope of TrueChain extends beyond luxury watches it can be applied to any industry vulnerable to counterfeiting, including fashion, electronics, pharmaceuticals, and automotive parts.

- End-to-End Traceability: Every movement of the product is recorded on blockchain.
- Scalability: Supports multiple categories of goods and can integrate with existing ERP systems.
- User Accessibility: The platform works on both web and mobile, providing real-time verification.
- Transparency: Every participant (manufacturer, supplier, retailer, consumer) can view verified product information.
- By combining blockchain, smart contracts, and QR technology, TrueChain ensures authenticity verification, strengthens supply chain security, and enhances industry reliability.

CHAPTER 5

Proposed System

The TrueChain web application provides a secure and transparent solution for verifying and tracking luxury products across the supply chain. It integrates blockchain technology to maintain tamper-proof product records and uses smart contracts to automate verification and ownership transfer. The system minimizes counterfeit risks, enhances transparency, and strengthens consumer trust by allowing real-time product authentication through unique QR codes. Built with React.js, Node.js, and PostgreSQL, the platform ensures a user-friendly interface for manufacturers, retailers, and consumers while maintaining strong data security and reliability.

5.1 Modules:

a. **User Registration & Authentication:** This module allows different users manufacturers, distributors, retailers, and consumers to register and log in securely. MetaMask integration ensures blockchain-based wallet authentication, while PostgreSQL stores encrypted user data and roles.

b. **Product Registration & QR Code Generation:** Manufacturers use this module to add new products to the blockchain. Each product is assigned a unique QR code linked to its blockchain record, containing details like product name, brand, and description. The QR code can be scanned anytime for verification.

c. **Product Verification & Consumer Access**: Consumers can scan the product's QR code through this module to check authenticity. The system fetches real-time blockchain data to confirm whether the product is genuine and displays its complete ownership history.

d. **Blockchain Integration & Data Storage**: This module handles smart contract operations and data management. Product and transaction data are stored immutably on the blockchain, while additional metadata (like product images or descriptions) is saved off-chain in PostgreSQL and IPFS for efficiency.

e. **Admin Dashboard & Monitoring**: The admin module helps monitor the overall system, manage users, and review blockchain transactions. It also provides analytics and visual reports, such as the number of verified products or transaction counts, using charts and tables.

f. **Smart Contract Management:** This module defines and executes the smart contracts that handle product registration, verification, and transaction validation automatically. It ensures transparency, immutability, and tamper-proof recordkeeping across the system.

5.2 Architecture Diagram:

The architecture shows how different users such as manufacturers, suppliers, retailers, and customers are connected through a secure blockchain network. The web application allows users to register products, verify authenticity, and manage records using smart contracts. These contracts handle key operations like product validation and data updates automatically. Product information is securely stored on the blockchain, while additional details are saved in a separate storage layer. The frontend connects with the blockchain through APIs, allowing smooth communication between users and the network. This design ensures that all product transactions are transparent, data remains tamper-proof, and customers can easily check product authenticity by scanning a QR code.

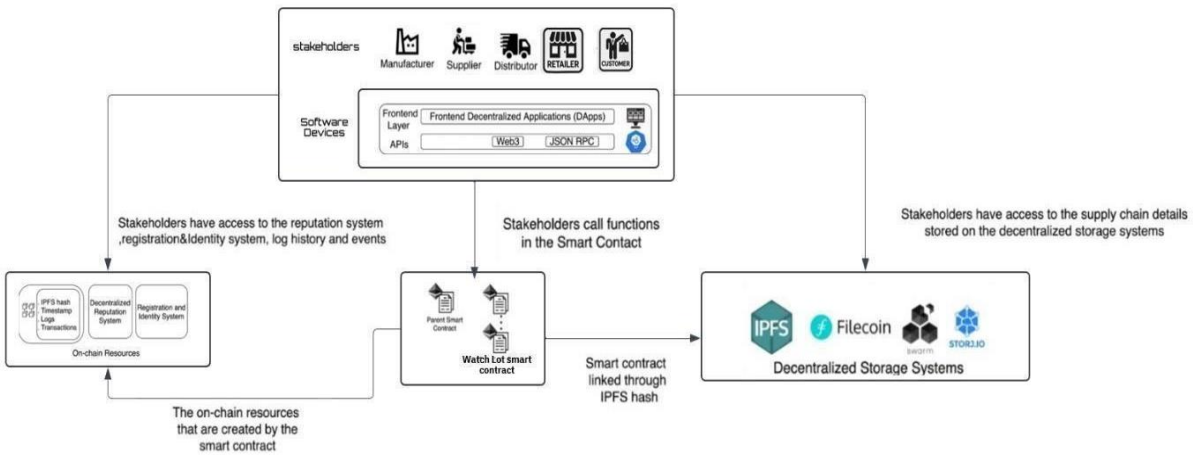


Figure 5.2 – Architecture Diagram

5.3 UML Diagrams:

A UML diagram is a diagram based on the UML (Unified Modelling Language) to visually represent a system along with its main actors, roles, actions, artifacts, or classes, to better understand, alter, maintain, or document information about the system.

5.3.1 DFD Diagram

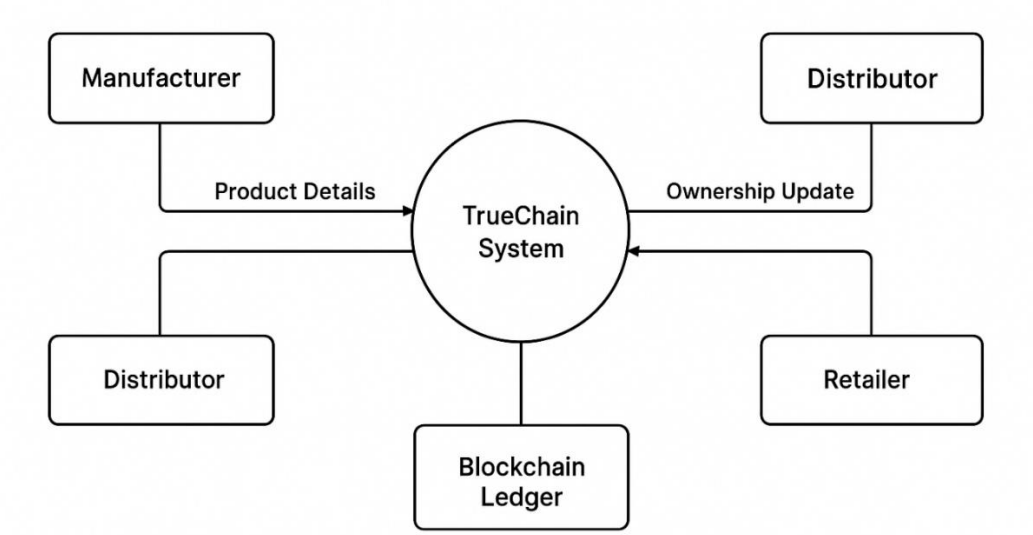


Figure 5.3.1.1 – DFD Level 0 Diagram

The Data Flow Diagram (DFD) Level 0 represents the overall workflow of the system at a high level. The manufacturer inputs product details into the system, which are securely recorded in the blockchain ledger. Distributors and retailers interact with the system to update and verify product ownership at each stage of the supply chain. The blockchain ledger ensures all transactions are immutable and transparent, enabling traceability and preventing counterfeit activities across the product lifecycle.

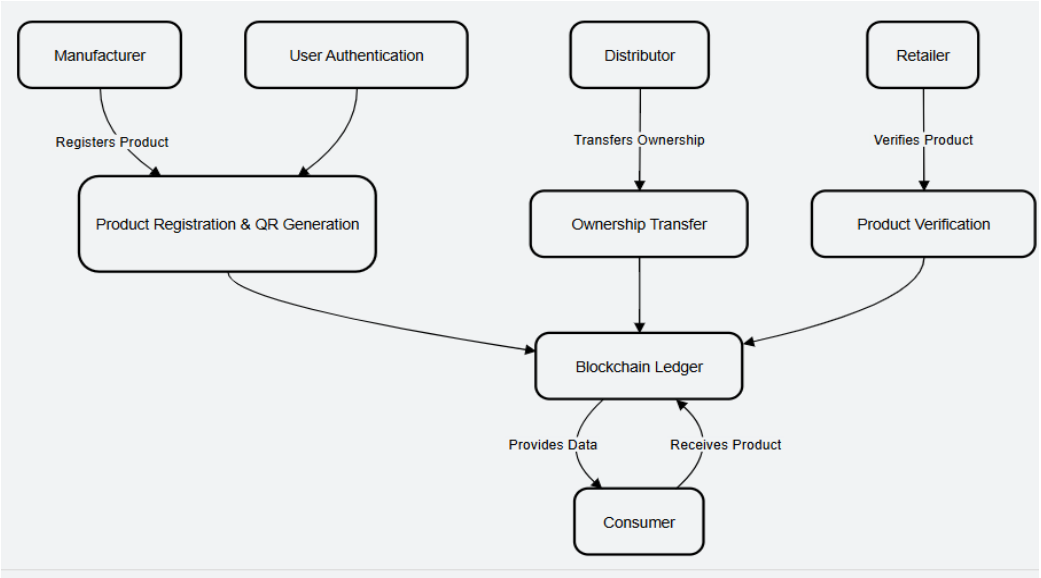


Figure 5.3.1.2 – DFD Level 1 Diagram

The Level 1 Data Flow Diagram illustrates the detailed workflow of the blockchain-based product authentication system. The Manufacturer registers products through the Product Registration & QR Generation process after successful User Authentication. Each registered product is recorded on the Blockchain Ledger, ensuring immutability. The Distributor transfers product ownership, and the Retailer verifies authenticity using the Product Verification process. Finally, the Consumer interacts with the blockchain to check product authenticity and history. This level ensures secure data flow, transparency, and traceability across all participants in the supply chain.

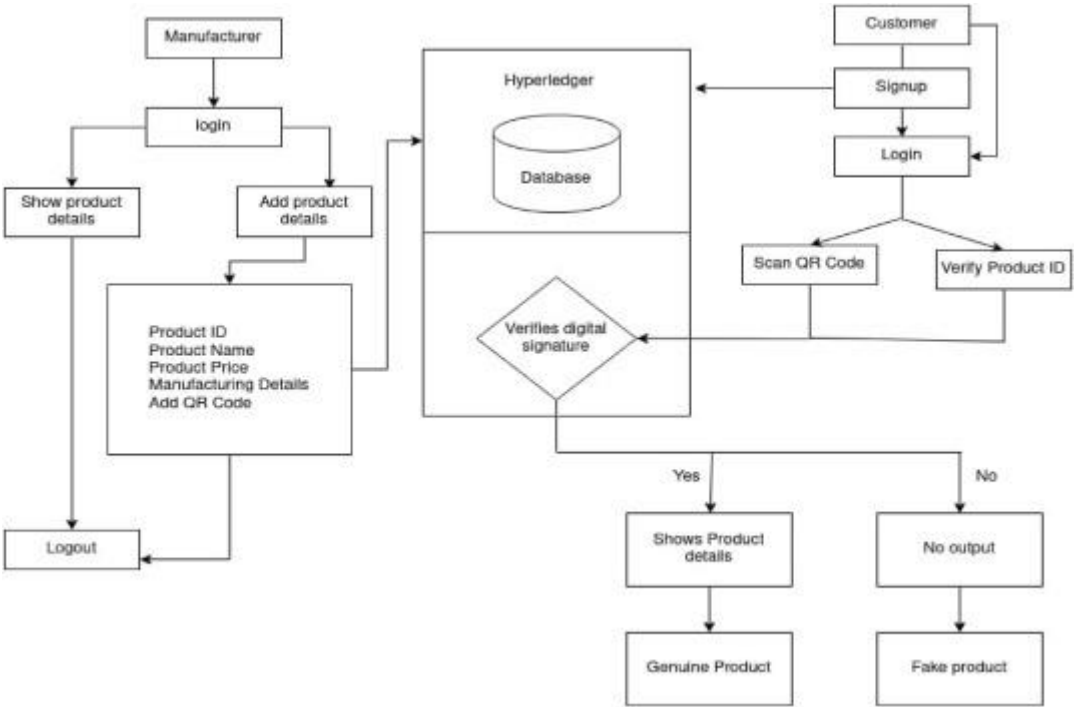


Figure 5.3.1.3 – DFD Level 2 Diagram

The Data Flow Diagram (DFD) Level 2 illustrates the detailed functioning of the system. The manufacturer logs in, registers products, and generates unique QR codes linked to blockchain entries. The blockchain module securely stores and verifies product information using smart contracts. Customers log in and scan QR codes to check product authenticity, and the system validates the data against blockchain records. Based on verification results, it displays whether the product is genuine or counterfeit, ensuring transparency and preventing fraud.

5.4 Use Case Diagram

This use case diagram shows how the TrueChain system connects all stakeholders in the supply chain manufacturer, distributor, retailer, and consumer. Each user signs up and logs in to the blockchain to perform actions like transferring products, and verifying authenticity. The manufacturer adds product details, which are then updated by the distributor and retailer as the product moves through the chain. Finally, the consumer scans the QR code to verify the product's authenticity. This ensures a secure, transparent, and traceable process from production to purchase.

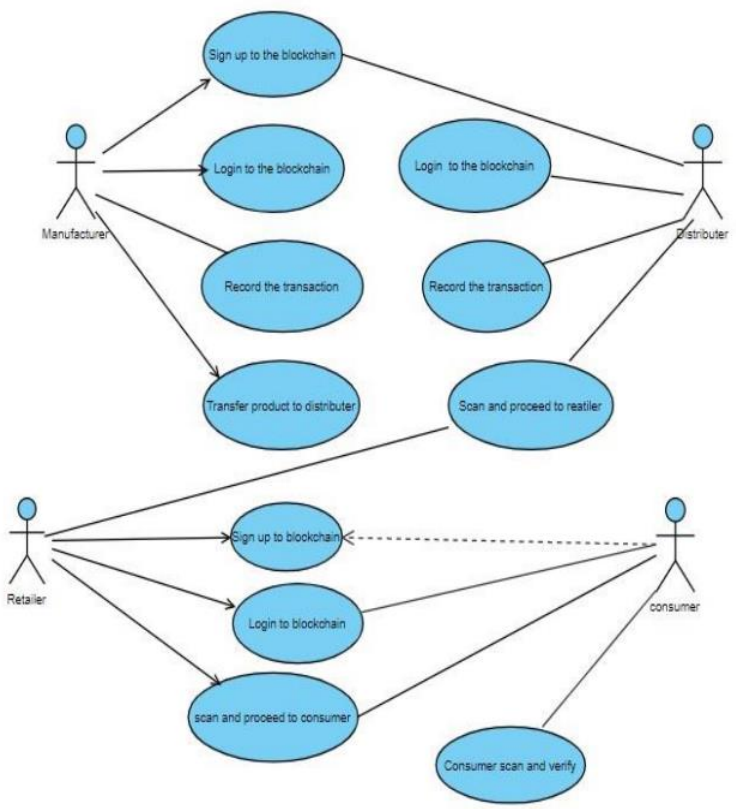


Figure 5.4 Use case Diagram

5.5 Sequence Diagram

This sequence diagram shows how the TrueChain system processes product registration, verification, and authentication using blockchain. The manufacturer enters product details through the frontend DApp, which connects to the MetaMask wallet for secure transaction signing. The signed data is sent to the smart contract, which stores product information on IPFS and records an immutable entry on the blockchain. Stakeholders like the distributor and retailer can scan and update the product status, while the customer verifies authenticity by scanning the QR code.

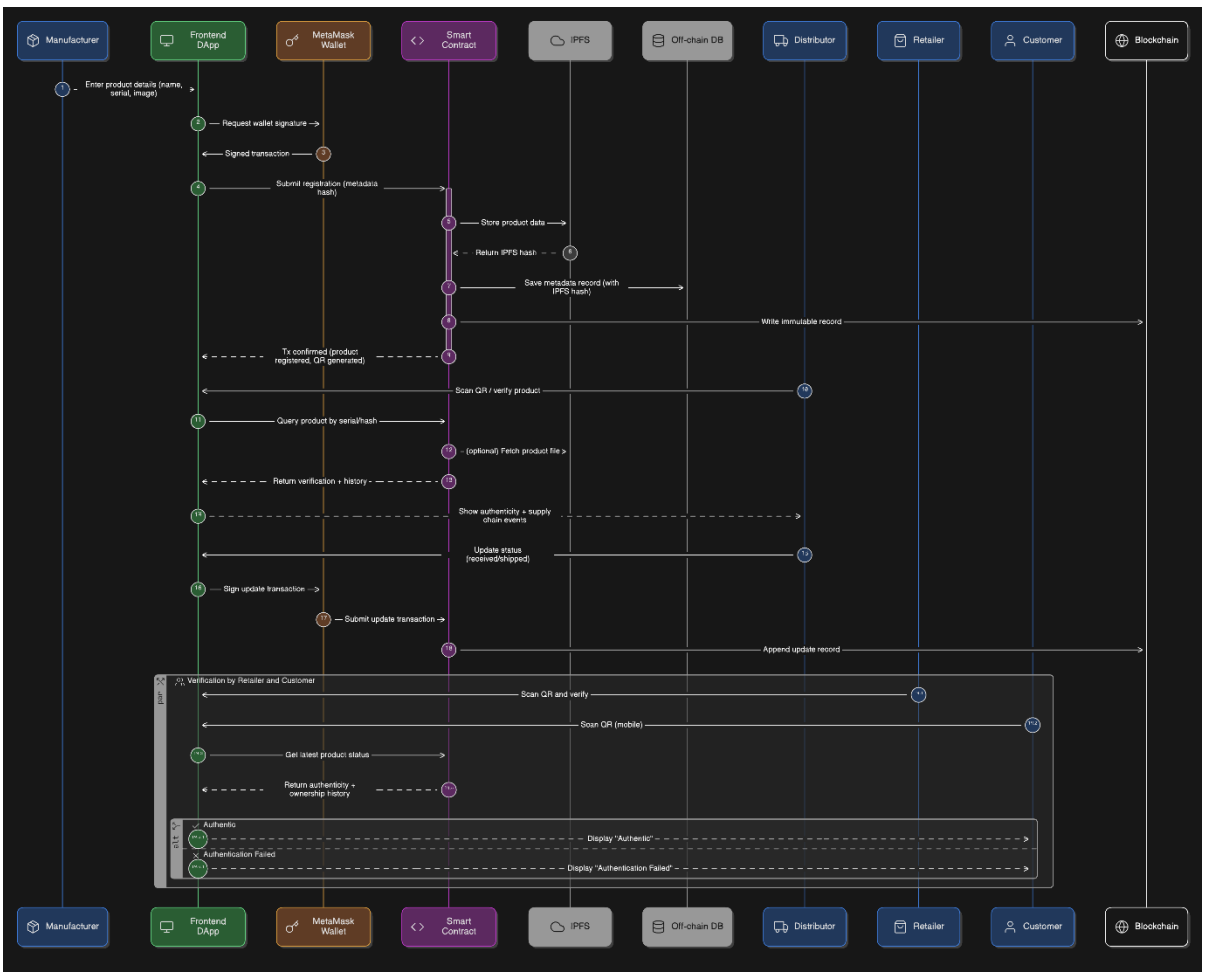


Figure 5.5 Sequence Diagram

5.6 Activity Diagram

An activity diagram shows how different stakeholders interact in the TrueChain system for product authentication. The administrator manages user accounts for the manufacturer, supplier, and retailer, who each update product details such as serial number, name, brand, and location on the Ethereum blockchain. As the product moves through the supply chain, its information and status are recorded at every stage.

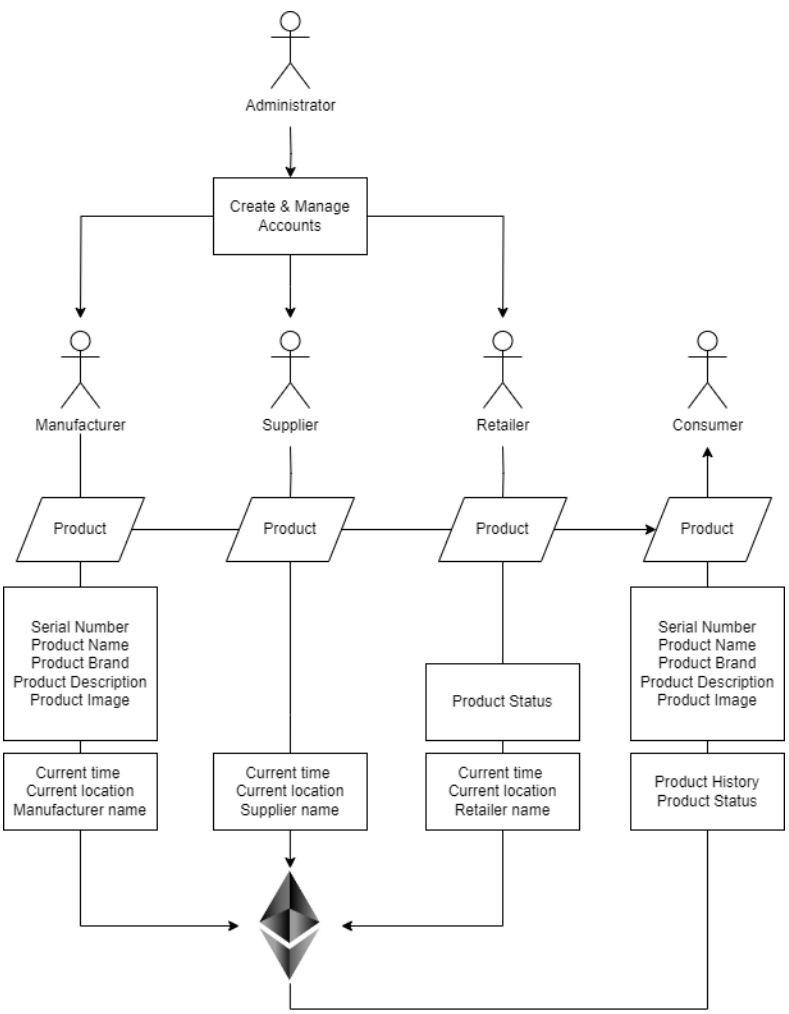
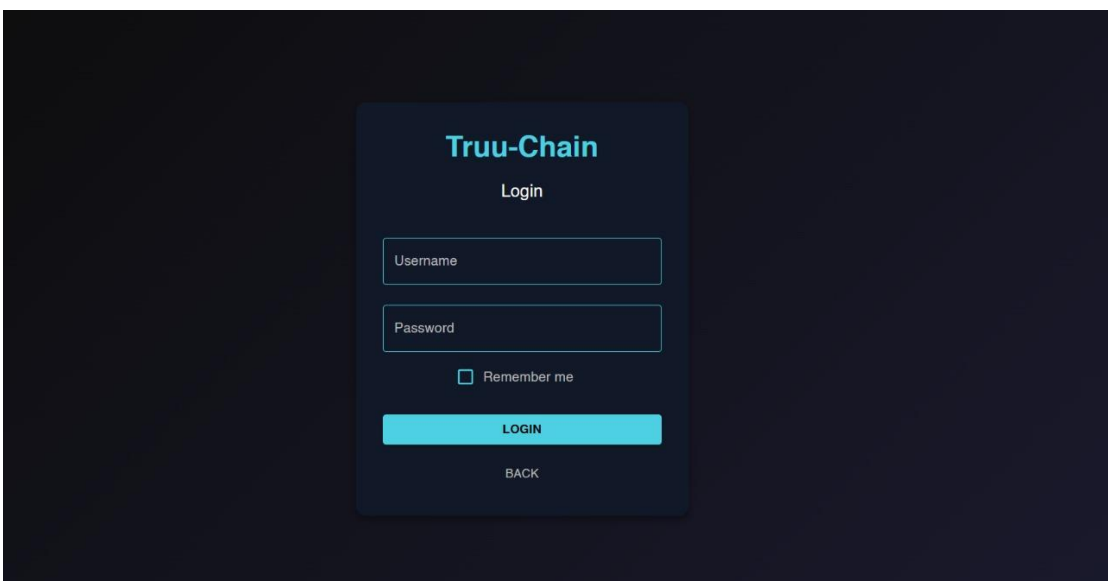


Figure 5.6 Activity Diagram

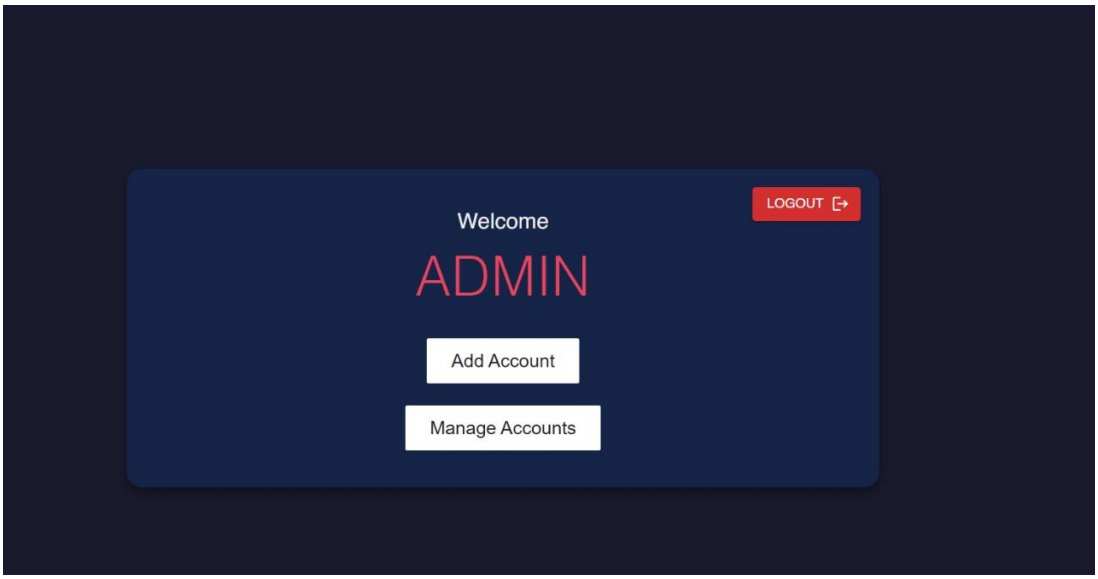
5.7 Implementation Details (Screenshots)



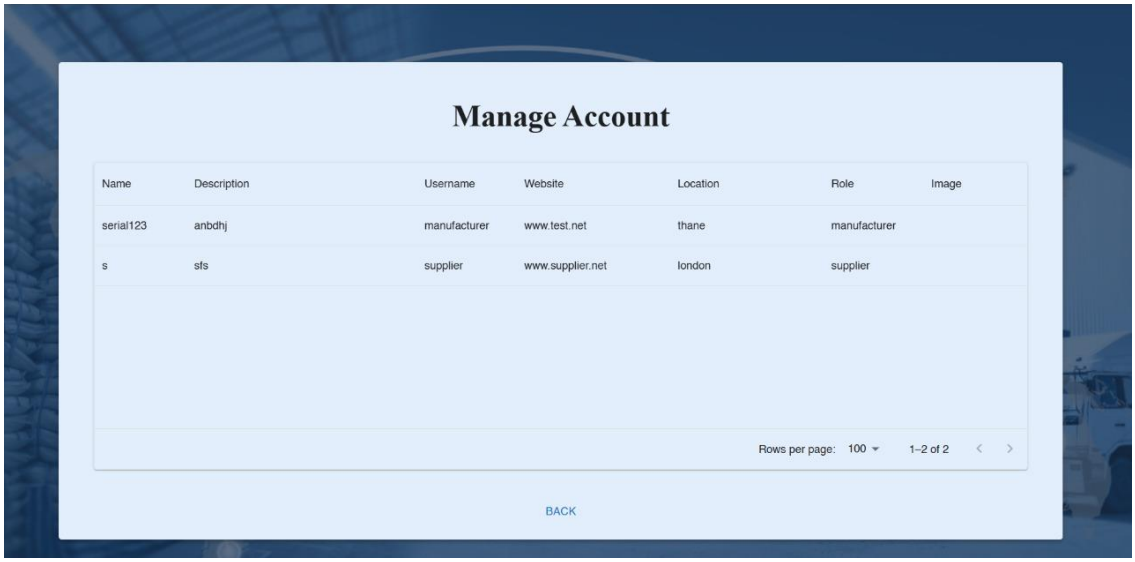
5.7.1 Home Page



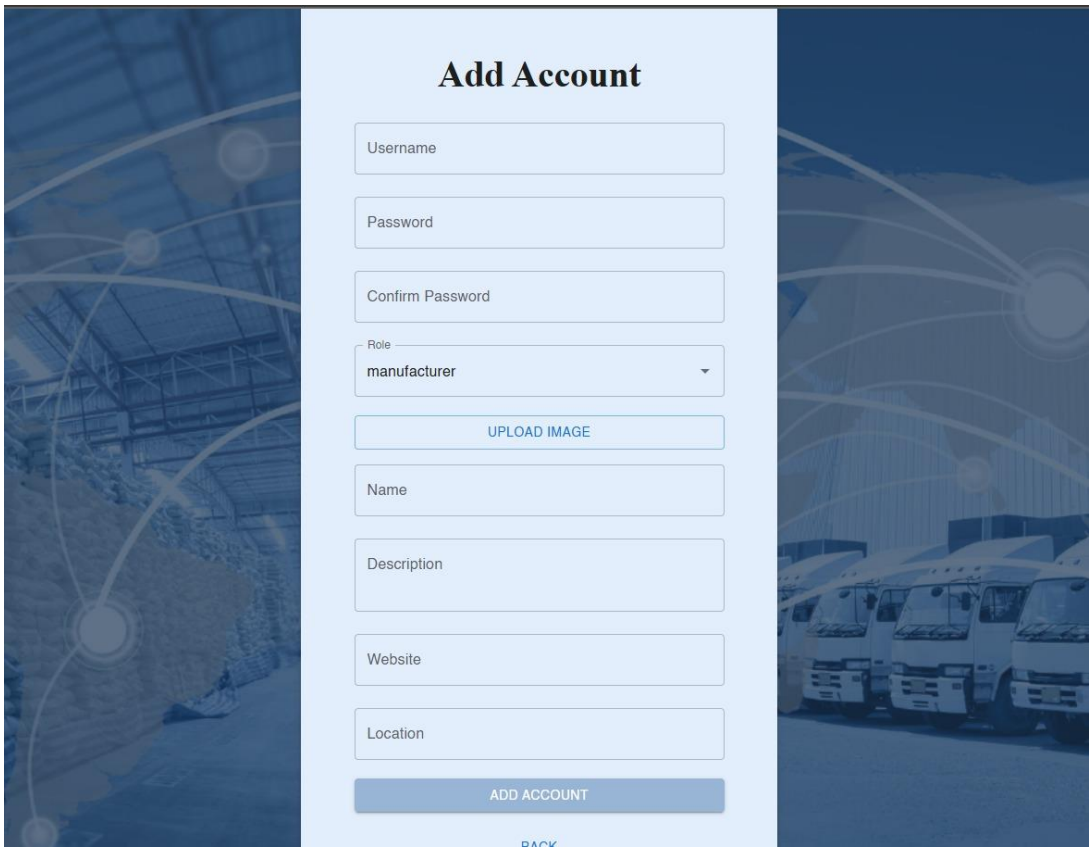
5.7.2 Login Page



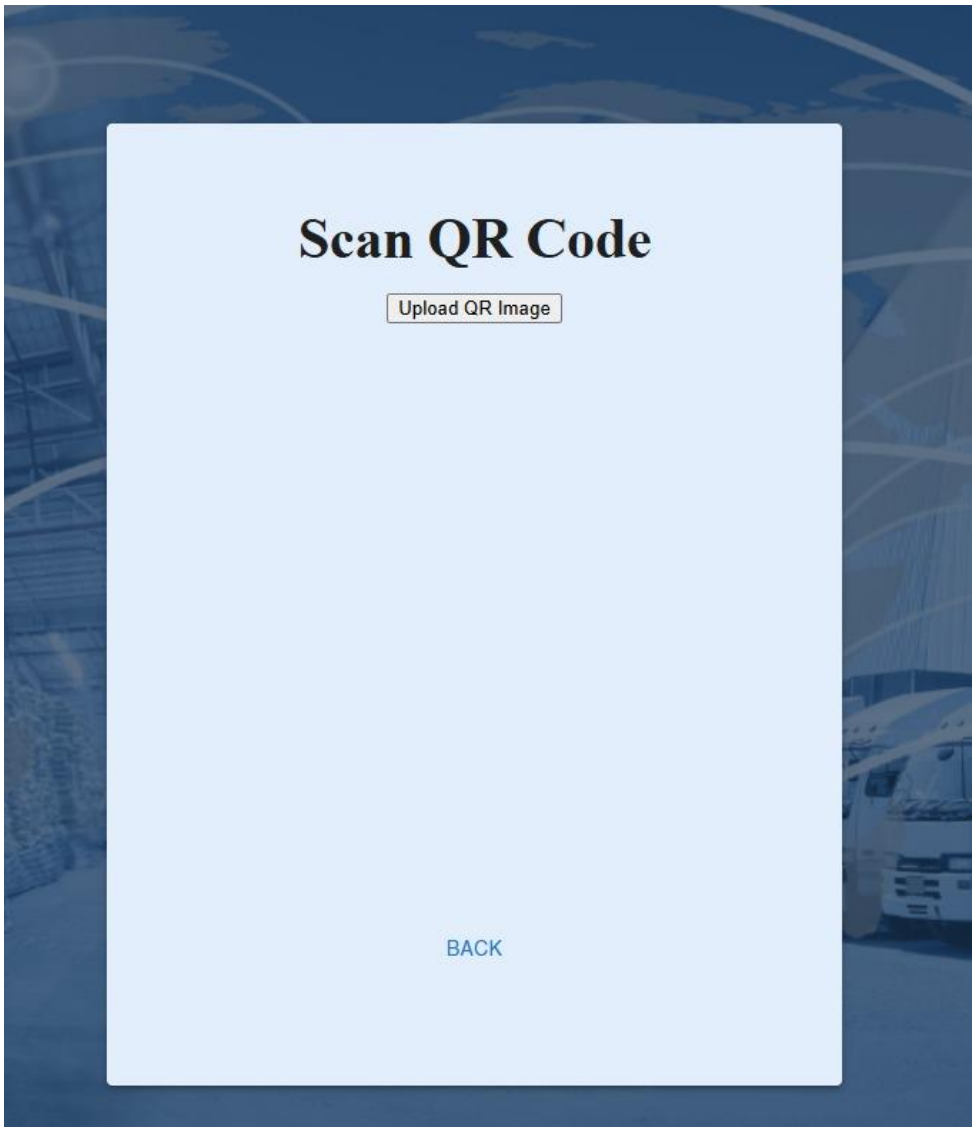
5.7.3 Admin Page



5.7.4 Manage Account Page



5.7.5 Add account / Product



5.7.6 Scan / Upload QR Image

CHAPTER 6

Experimental Setup

6.1 Technology Stack

The TrueChain system is built on a secure and scalable architecture that combines Node.js, Express.js, React.js, and PostgreSQL with Solidity-based smart contracts deployed on the Ethereum blockchain. The platform uses QR code technology for product identification and blockchain integration to ensure transparency, authenticity, and traceability across the supply chain.

6.1.1 Frontend (UI):

The frontend application provides an intuitive and user-friendly interface for manufacturers, distributors, retailers, and consumers to interact with the blockchain securely.

a. Framework – React.js:

Used to develop a responsive, component-based user interface for pages like Product Registration, Verification, and Dashboard. React's virtual DOM enables fast rendering and smooth navigation.

b. Styling – Tailwind CSS:

Tailwind CSS provides a modern, consistent design system with reusable utility classes, ensuring responsive layouts and clean aesthetics across devices.

c. Routing – React Router:

Enables seamless navigation between different modules like Login, Register Product, and Verify Product without full page reloads, improving performance.

d. API Communication – Axios:

Used for asynchronous communication with backend APIs to handle product registration, QR code generation, and blockchain verification requests securely.

6.1.2 Backend (Server):

The backend serves as the main processing layer that connects the frontend with blockchain smart contracts and the PostgreSQL database.

a. Runtime – Node.js:

Provides an efficient runtime environment for executing backend logic, managing APIs, and handling data flow between the blockchain and the frontend.

b. Framework – Express.js:

Simplifies server-side operations by managing routing, API endpoints, and communication with blockchain and database modules, ensuring scalability and security.

c. Database – PostgreSQL:

A relational database used to store off-chain product metadata, user profiles, and QR mappings. It ensures structured, secure, and high-performance data handling.

d. Authentication – MetaMask Integration:

Allows users to log in using Ethereum wallets instead of traditional credentials, providing decentralized authentication and secure blockchain interaction.

e. API Management – RESTful APIs:

Express-based RESTful APIs are used for handling operations such as product registration, verification, and transaction history retrieval.

6.1.3 Blockchain Layer (Product Authentication & Tracking):

This layer ensures immutable, transparent, and decentralized management of product data and transactions across the supply chain.

a. Smart Contract Language – Solidity:

Used to write smart contracts that define product registration, authenticity checks, and verification logic on the Ethereum blockchain.

b. Development Environment – Hardhat:

Provides a local blockchain testing environment to compile, test, and deploy smart contracts efficiently before deployment on Ethereum networks.

c. Wallet Integration – MetaMask:

Used for secure transaction signing and wallet-based authentication, enabling users to interact directly with the blockchain through their Ethereum wallets.

d. Blockchain Network – Ethereum Testnet:

The system operates on an Ethereum test network to validate product authenticity and ownership securely using decentralized transactions.

6.1.4 QR Code and Data Storage Layer:

This layer handles product identification, data linking, and secure off-chain data storage.

a. QR Code Library – JavaScript QR Generator:

Generates unique QR codes for each registered product. These codes link blockchain transaction hashes with product data for easy authenticity verification.

b. Off-Chain Storage – PostgreSQL:

Stores non-sensitive metadata such as product names, descriptions, timestamps, and QR mappings for efficient retrieval.

c. Workflow:

When a product is registered, its key details are stored on the blockchain, and a QR code is generated linking it to off-chain records, enabling easy product verification by scanning.

6.1.5 Containerization and Deployment:

This layer ensures smooth deployment, scalability, and consistent performance across environments.

a. Technology – Docker:

Used to containerize the entire application, including the backend, frontend, and blockchain integration, ensuring portability and consistent execution during development and testing.

b. Workflow:

Each module (frontend, backend, blockchain node) runs in separate containers that communicate seamlessly, reducing configuration errors and simplifying deployment

6.2 Software and Hardware Setup

- **Software Requirements**

b. Node.js

c. Express.js

d. Solidity

e. Hardhat

f. PostgreSQL

g. QR Code Library

h. Docker

- **Hardware Requirements**

1. Processor: Minimum Intel Core i3 or equivalent.
2. Memory (RAM): Minimum 4 GB for stable execution and testing.
3. Storage: At least 512 GB SSD for faster data access and blockchain deployments.
4. Operating System: Windows 10 or above / macOS / Linux for cross-platform compatibility.
5. Internet Connection: Required for blockchain network access, Hardhat testnet operation, and real-time database communication

Chapter 7

Project Plan

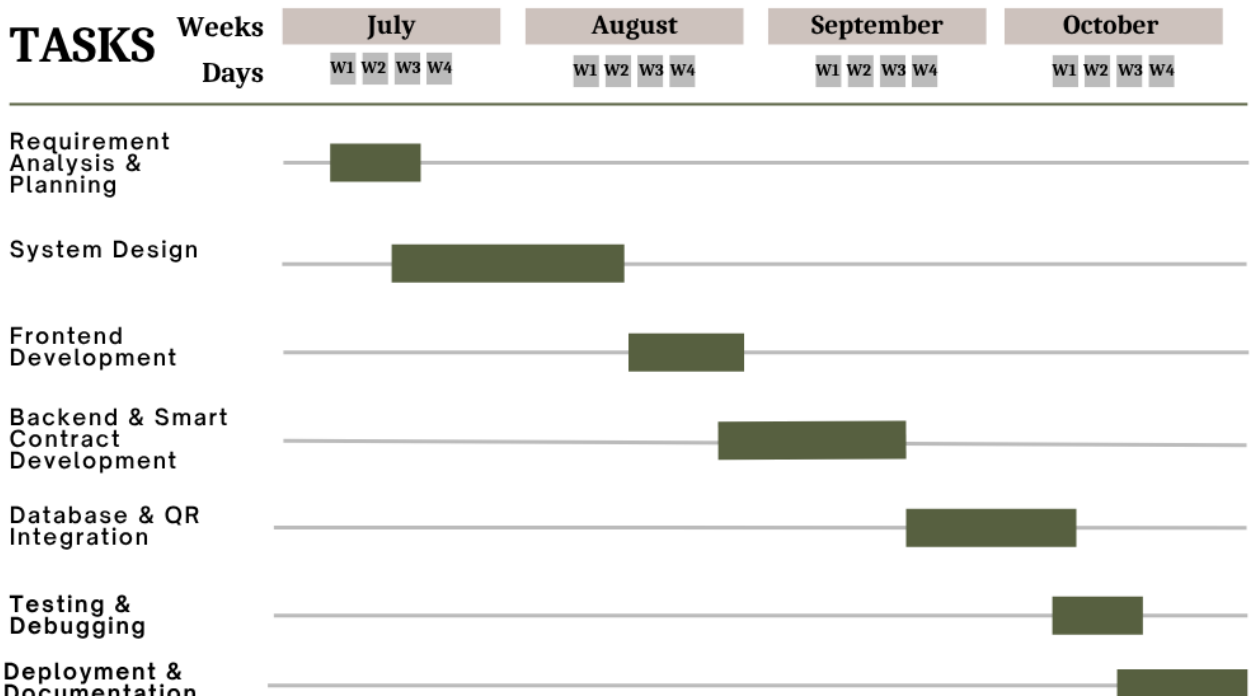


Figure 7.1 – Gantt Chart

The Gantt chart illustrates the project timeline for TrueChain from July to October, covering key development phases. The process started with requirement analysis and planning in early July, followed by system design through late July and August. Frontend development was carried out during mid-August, while backend and smart contract development continued into September. Database and QR integration were completed by early October, followed by testing and debugging to ensure proper functionality.

CHAPTER 8

Expected Outcome

The expected outcomes of TrueChain, a blockchain-based product authentication and tracking system using QR code technology, are highly impactful and aim to improve transparency, security, and trust within the luxury goods industry. The project focuses on eliminating counterfeit products, enhancing consumer confidence, and strengthening brand protection across the global supply chain. Some key expected achievements include:

1. **Secure and Transparent Product Authentication:** TrueChain ensures that every product registered on the blockchain is uniquely identifiable and verifiable. Each transaction such as manufacturing, distribution, and sale is recorded immutably, preventing tampering or duplication and ensuring complete authenticity verification for consumers.

2. **Decentralized and Tamper-Proof Data Storage:** By leveraging blockchain technology, the system maintains an immutable ledger of product records. This decentralized architecture eliminates the need for third-party verification while ensuring that data remains secure, transparent, and resistant to unauthorized modification.

3. **Real-Time Verification through QR Codes:** Consumers can scan product-specific QR codes to instantly check authenticity and access detailed product information, including origin, manufacturing date, and transaction history. This feature helps build customer trust and discourages the purchase of counterfeit goods.

4. **Real-Time Verification through QR Codes:** Consumers can scan product-specific QR codes to instantly check authenticity and access detailed product information, including origin, manufacturing date, and transaction history.

5. **Brand Protection and Supply Chain Transparency:** Manufacturers and retailers can track products across every stage of the supply chain, from production to sale. This visibility helps brands detect and prevent counterfeit distribution while maintaining transparency with suppliers and customers.

6. **Automated Smart Contract Operations:** Smart contracts handle key processes such as product registration, verification, and data validation automatically. This reduces manual errors, speeds up authentication, and ensures that all actions are traceable and verifiable on the blockchain.

7. **Improved Consumer Trust and Market Credibility:** By providing real-time proof of authenticity, TrueChain enhances consumer confidence in luxury brands. It helps maintain brand reputation, boosts customer loyalty, and promotes ethical consumption in global markets.

8. **Scalability and Future Integration:** The modular design and blockchain-based architecture make TrueChain easily adaptable for various industries beyond luxury goods.

In conclusion, TrueChain ensures secure and transparent product authentication using blockchain and QR code technology. It helps eliminate counterfeit goods, protects brand integrity, and builds consumer trust. The system's decentralized design promotes accountability and can be extended to various industries for reliable product verification.

Chapter 9
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